REPORT NO. 4A: REVIEW ON UML DRAWING TOOLS

COURSE CODE: CSE 314
COURSE TITLE: OBJECT ORIENTED ANALYSIS AND DESIGN LABORATORY

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Abstract

This laboratory activity focuses on exploring and evaluating different UML (Unified Modeling Language) drawing tools through collaborative teamwork. A group of team members was formed, and each member was assigned a specific UML tool to research and analyze, including Lucidchart, Draw.io, PlantUML, Diagrams.net, and SmartDraw. The objective of the experiment was to understand the features, advantages, and limitations of commonly used UML diagramming tools and to determine the most suitable tool for team-based project work. Each member prepared a brief review of their assigned tool, after which the team compared all tools based on usability, cost, collaboration features, and suitability for academic projects. The activity emphasized teamwork, communication, research skills, and decision-making in selecting appropriate software development tools. This exercise provides a foundational understanding of UML diagram creation and tool selection, which are essential for object-oriented analysis, system design, and effective project management in software development.

1. OBJECTIVE

This laboratory experiment focuses on exploring and evaluating different UML (Unified Modeling Language) drawing tools through collaborative teamwork. The activity introduces students to various diagramming tools used in Object-Oriented Analysis and Design (OOAD) and helps them understand how to select appropriate tools for project development. By researching, comparing, and presenting multiple UML tools, students gain practical exposure to teamwork, task distribution, and tool selection, which are essential in software development environments.

1.1 Objectives

The objectives of this laboratory activity are as follows:

- To form a project team and assign different UML tools to each member.
- To develop teamwork, research, and communication skills among team members.
- To explore the features and functionalities of commonly used UML drawing tools.
- To identify the advantages and limitations of each UML tool.
- To compare different tools based on usability, cost, collaboration, and platform support.
- To select the most suitable UML tool for team project work.
- To document findings and prepare a structured team report.
- To gain practical understanding of UML tools used in Object-Oriented Analysis and Design (OOAD).

1.2 Expected Outcomes

At the end of this laboratory activity, the following outcomes are expected:

- Successful formation of a project team with task distribution among members.
- Completion of individual research on assigned UML tools.
- Compilation of a comparative report on five UML drawing tools.
- Identification of advantages and limitations of each tool.
- Selection and justification of the most suitable UML tool for the team project.
- Improved understanding of UML diagram creation and tool usage.
- Enhanced teamwork, collaboration, and report-writing skills.
- Preparation for future software design and OOAD laboratory activities.

2. EQUIPMENT USED

This chapter describes the equipment and resources used to conduct the laboratory activity on exploring UML drawing tools and preparing a comparative team report. The listed equipment supported research, diagram creation, collaboration, communication, and documentation throughout the experiment.

The following equipment and resources were used to complete the activity:

- Computer or laptop with internet access for researching UML tools and creating diagrams.
- Web-based UML drawing tools such as Draw.io, Lucidchart, PlantUML, Diagrams.net, and SmartDraw.
- Diagram creation software (Draw.io desktop or browser version) for preparing sample UML diagrams.

- Microsoft Word or Google Docs for writing and formatting the laboratory report.
- Presentation and collaboration platforms such as Google Drive for file sharing and storage.
- Trello board for task management and tracking team progress.
- Discord (or similar communication platform) for team discussion and coordination.
- Smartphone or screenshot tools for capturing evidence of team collaboration and contributions.
- Cloud storage services (Google Drive/Dropbox) for backup and safe storage of documents and diagrams.

3. PROCEDURE

This chapter describes the step-by-step procedure followed during the laboratory activity. The team collaboratively explored different UML tools, managed tasks using Trello, communicated through Discord, and finally selected Draw.io as the most suitable UML tool.

3.1 Phase 1: Tool Assignment and Task Management

At the beginning of the lab session, five UML tools were assigned to five team members. Each member was responsible for researching one tool and preparing a short report section.

The assigned tools were:

- Lucidchart
- Draw.io (diagrams.net)
- PlantUML

- SmartDraw
- Creately

To manage tasks, the team used Trello with three sections:

- To Do
- Doing
- Testing/Review

Team communication was handled through Discord where members shared files, updates, and task assignments.

Trello Activity

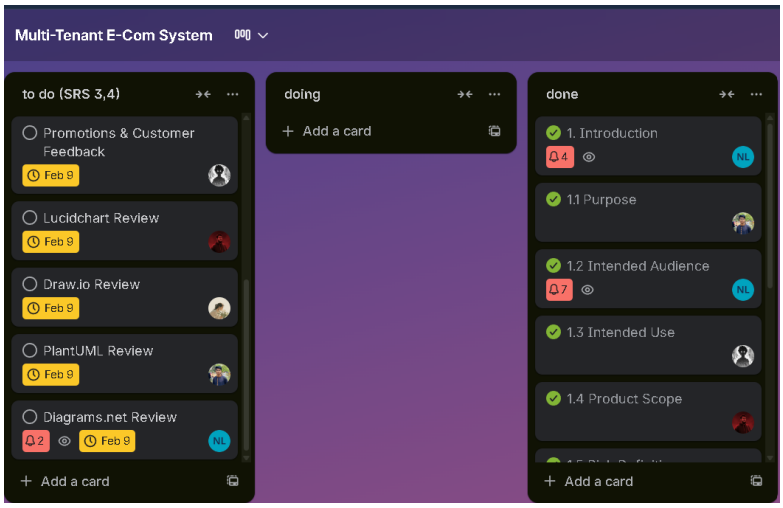

Figure 3.1: Trello board showing To-Do, Doing, and Testing sections

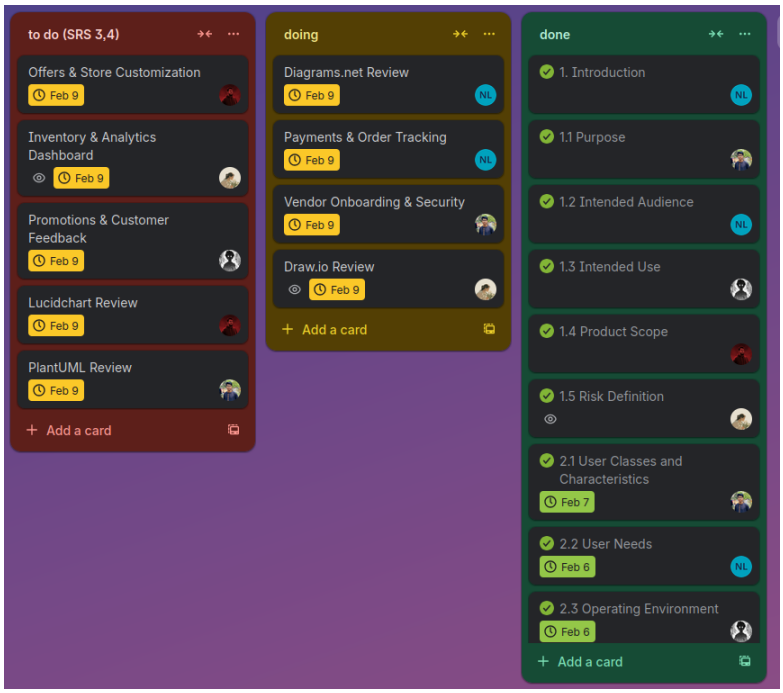

Figure 3.2: Trello activity notification

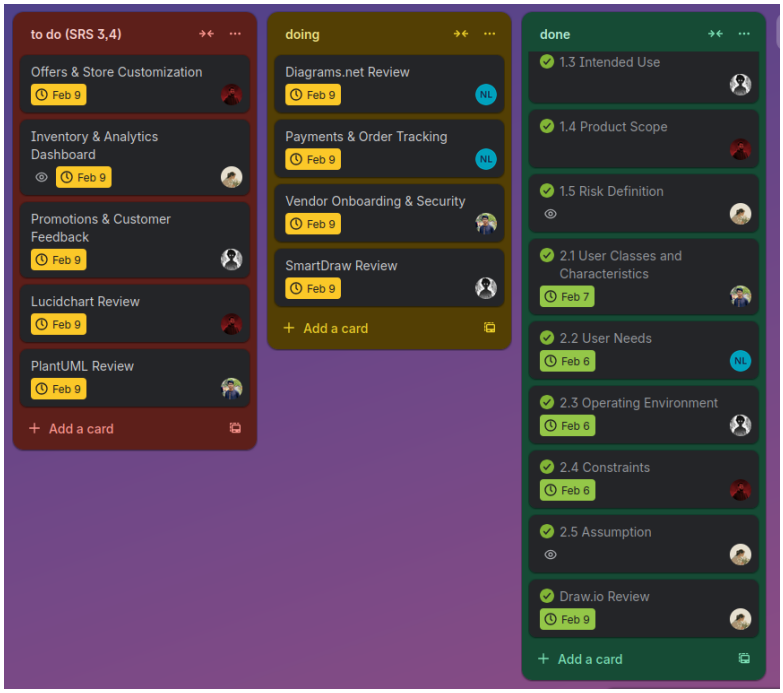

Figure 3.3: Trello task update notification

Discord Communication

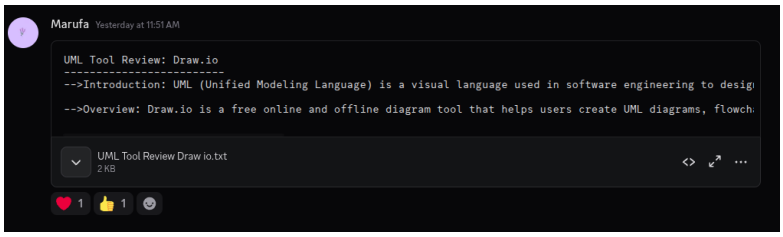

Figure 3.4: File sharing in Discord

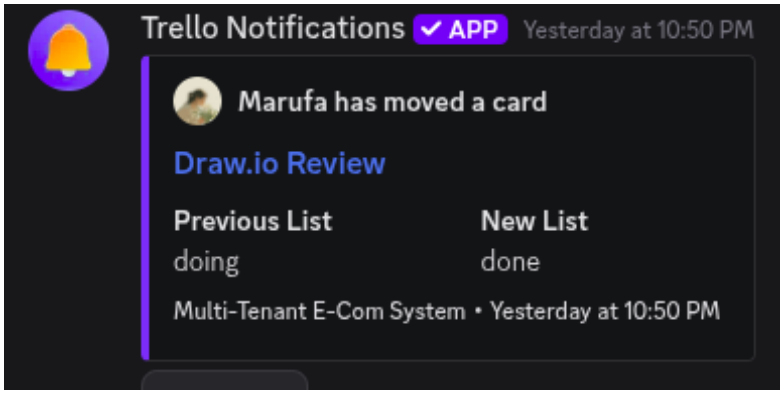

Figure 3.5: Task assignment discussion in Discord

3.2 Phase 2: Tool Research and Exploration

Each member explored their assigned UML tool by:

- Studying features and interface
- Creating sample UML diagrams
- Identifying advantages and limitations

After completing the research, the team compared all tools.

3.3 Phase 3: Final Tool Selection

After discussion and comparison, the team selected **Draw.io** as the final UML tool.

Reasons for choosing Draw.io:

- Free and easy to use
- Web-based and desktop support
- Drag-and-drop interface
- Supports all UML diagrams
- Easy export and sharing

3.4 Phase 4: Report Preparation and Submission

All members completed their report sections. The final report was compiled, reviewed, and submitted with screenshots of Trello and Discord activities as proof of collaboration.

4. DATA

This chapter presents the data collected during the laboratory activity along with the verification details for each task.

UML Tool	Assigned Member	Status	Verified By
Lucidchart	Rohan	Completed	Team
Draw.io	Marufa	Completed	Team
PlantUML	Shihab	Completed	Team
Diagrams.net	Lineya	Completed	Team
SmartDraw	Taskif	Completed	Team

Table: UML Tool Review Task Data

5. OBSERVATIONS AND DISCUSSIONS

This chapter discusses the observations made during the laboratory activity and analyzes the effectiveness of using collaborative tools and UML diagram software for completing the team-based assignment.

- Assigning different UML tools to each team member allowed the team to explore a wide range of diagramming platforms and understand their unique features.
- The use of Trello provided a clear visual workflow with *To Do*, *Doing*, and *Done* columns, which helped track progress and manage tasks efficiently.
- Discord communication enabled quick discussion, task clarification, and file sharing among team members, improving coordination and teamwork.
- Independent research on assigned tools helped develop individual understanding while still contributing to the team report.
- Creating sample UML diagrams in each tool allowed members to compare usability, features, and flexibility in real scenarios.
- Screenshot evidence from Trello and Discord served as proof of participation, task completion, and collaboration.
- During the comparison phase, Draw.io was observed to be the most suitable tool due to its free access, simple interface, wide template support, and easy export options.
- The activity demonstrated how collaborative platforms and diagramming tools can be integrated effectively in software engineering workflows.
- Instructor feedback highlighted the importance of documentation, tool comparison, and proper communication in team-based software projects.

6. CONCLUSIONS

This laboratory activity successfully demonstrated the process of exploring and evaluating different UML drawing tools through structured teamwork and collaborative platforms. The experiment helped students understand how software tools can be researched, compared, and selected based on usability, features, and project requirements.

The use of Trello for task management and Discord for communication provided an organized workflow for assigning responsibilities, tracking progress, and sharing updates. These tools improved coordination among team members and ensured that all tasks were completed within the given timeframe.

Through individual research and group discussion, the team was able to compare multiple UML tools and identify Draw.io as the most suitable option for academic and collaborative use due to its simplicity, accessibility, and free availability. The activity also reinforced the importance of documentation, communication, and evidence-based tool selection in software engineering projects.

Overall, this laboratory session established a strong foundation for future Object-Oriented Analysis and Design activities by introducing collaborative workflows, UML tool evaluation, and structured reporting practices that are essential in real-world software development environments.